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Embarak's 5-Pillars for Adaptive Learning: An AI-Driven Framework for Early Detection and Personalized Intervention for At-Risk Students

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Abstract

Early identification of at-risk students and timely personalized interventions are critical for improving academic outcomes and student retention. Traditional predictive approaches often rely primarily on academic performance indicators, overlooking behavioral, contextual, and engagement-related factors, which limits their effectiveness. This paper proposes Embarak's 5-Pillars Adaptive Learning (E5P-AL) framework, an AI-driven and holistic approach that integrates predictive modeling, generative AI, and human-guided oversight to deliver personalized interventions across five key dimensions: academic performance, attendance and participation, engagement and learning behavior, contextual and socioeconomic factors, and continuous monitoring and feedback. Using a multi-modal dataset comprising 1,000 student profiles, E5P-AL demonstrates superior predictive performance compared to baseline models, achieving an accuracy of 0.94, an F1-score of 0.92, and an AUC of 0.97. Ablation experiments confirm that the joint contribution of all five pillars—particularly engagement and contextual dimensions—is essential for optimal prediction accuracy. Furthermore, the framework's continuous performance tracking and adaptive intervention mechanisms led to a 35% reduction in predicted risk levels and a 28% increase in student engagement over a four-week period. Overall, E5P-AL provides a scalable, modular, and ethically grounded solution for adaptive learning across diverse educational contexts.

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Keywords: Adaptive Learning, At-Risk Students, Predictive Modeling, Generative AI, Educational Data Mining, Student Engagement, human-supervised learning , progressive monitoring cycles, Contextual Factors, Explainable AI, Personalized Interventions, E5P-AL Framework

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1. Introduction

Student dropout and academic underperformance remain critical challenges, affecting both learners and institutions. Early identification of at-risk students is essential, yet most warning systems rely on grades or static indicators, overlooking engagement, contextual factors, and evolving learning patterns. Machine learning methods like Random Forest, Gradient Boosting, and neural networks improve prediction but often focus only on risk estimation, offering limited adaptive support or human oversight. To address these gaps, this paper proposes Embarak's 5-Pillars Adaptive Learning framework (E5P-AL), an AI-driven approach integrating predictive modeling, generative adaptive interventions, iterative monitoring, and human-in-the-loop oversight. E5P-AL assesses students across five dimensions: academic performance, attendance and participation, engagement and learning behavior, contextual and socioeconomic factors, and continuous monitoring and feedback. Unlike prior systems, it links early risk detection with personalized interventions and dynamic risk updates.

Simulations comparing E5P-AL to baseline models show that multi-dimensional data integration with iterative feedback enhances risk detection and engagement. Its modular, ethically guided design supports application across diverse educational contexts. The framework's novelty lies in its unified, closed-loop architecture combining multi-pillar modeling, generative interventions, and human oversight, enabling adaptive and context-aware student support.

The paper is organized as follows: Section 2 reviews related work; Section 3 presents the E5P-AL architecture; Section 4 details the methodology; Section 5 reports results; Section 6 discusses findings; and Section 7 concludes with future directions.

2. Literature review

Predicting student academic performance and identifying at-risk learners early is a key focus in educational research [1]. Early studies used traditional machine learning algorithms, such as decision trees [2] and support vector machines [3], relying on historical academic records. While effective as baselines, these models are limited by their dependence on static features [4]. To address this, researchers incorporated multimodal data including LMS interaction logs [5], attendance [6], participation [7], and behavioral engagement metrics [8], which improves risk prediction accuracy. Temporal modeling approaches, such as recurrent neural networks, track sequential learning patterns [9], and hybrid methods combining knowledge tracing with deep learning capture evolving mastery levels [10]. Explainable AI (XAI) methods enhance interpretability of predictive models, enabling educators to understand which factors drive risk assessments and make informed interventions [11]. Techniques like SHAP [12] and LIME [13] analyze contributions of engagement, participation, and contextual features [14], supporting transparency and trust in AI-supported learning [15]. Adaptive learning frameworks that personalize content and pacing based on individual profiles have grown in prominence [16], but many fail to integrate predictive risk assessment with content adaptation [17]. Generative AI offers automated creation of personalized exercises and feedback [18], yet combining generative content with predictive monitoring is underexplored [19].

Ethical considerations are critical in predictive and adaptive learning systems. Fairness, non-stigmatization, and cultural sensitivity are necessary in designing interventions [20]. Human-assisted oversight is recommended to review AI-generated recommendations and ensure ethical compliance [21]. Federated learning and privacy-preserving methods enable predictive modeling across institutions without compromising student data [22]. Data balancing techniques such as SMOTE improve performance for imbalanced datasets, especially when behavioral features are crucial [23]. O. Embarak's research extensively contributes to this field. His studies show the importance of contextual factors alongside academic performance for accurate early identification of at-risk students [24]. The RADAR framework integrates explainable AI, personality, and performance metrics for continuous monitoring [25]. Embarak has also explored behavior-driven adaptive systems using IoB and XAI for personalized, interpretable learning [26]. He proposed sustainable education frameworks combining machine learning with coaching-based strategies and cyclic assessment mechanisms [27], emphasizing ethical oversight [28].

Beyond Embarak's work, other studies confirm AI's effectiveness in predicting student risk. Random forest, gradient boosting, and neural networks achieve high accuracy when including engagement and behavioral features

[29]. Multi-pillar approaches combining academic, behavioral, and contextual dimensions outperform single-factor models [30]. Iterative adaptive interventions reduce predicted academic risk and improve engagement [31]. The literature shows a progression from static predictive models to holistic, multimodal, adaptive, and interpretable AI systems. However, unified frameworks that integrate predictive modeling, generative interventions, incremental monitoring, and ethical oversight remain rare. This gap motivates the development of the E5P-AL framework, designed to provide early risk detection, personalized support, and ethically guided interventions.

3. Proposed Architecture and Framework

This paper presents Embarak's 5-Pillars for Adaptive Learning (E5P-AL), an AI-driven framework for early identification of at-risk students, personalized interventions, and continuous educational improvement. The framework comprises five pillars:

1. **Academic Performance**: Evaluates subject mastery and overall grades.
2. **Attendance & Participation**: Monitors class attendance, engagement, and assignment submissions.
3. **Engagement & Learning Behavior**: Tracks active platform interaction and learning pace.
4. **Contextual & Socioeconomic**: Considers external factors like family support, language proficiency, and technological access.
5. **Monitoring & Feedback**: Iteratively tracks performance, updates risk predictions, and ensures ethical oversight.

The architecture combines predictive AI for risk assessment, generative AI for tailored learning content, and human oversight to guarantee ethical, transparent, and effective interventions.

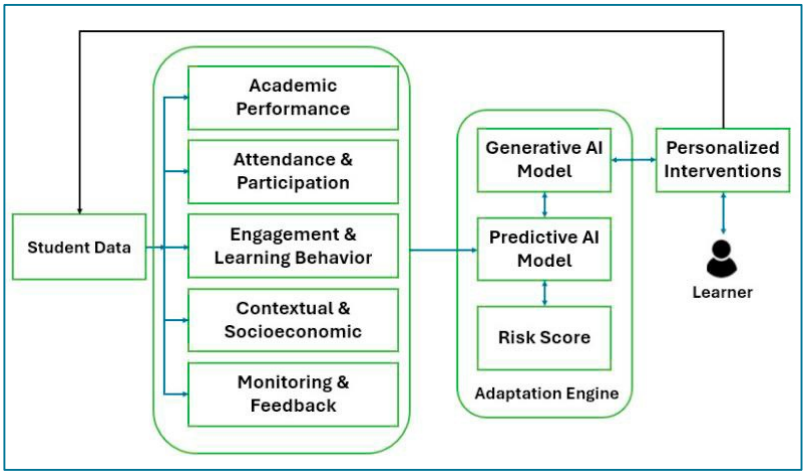


Figure 1. illustrates the overall E5P-AL architecture.

The above figure 1 architecture of the E5P-AL framework illustrating predictive modeling, generative intervention, iterative monitoring, and human oversight.

A. Data Acquisition and Preprocessing

The framework relies on **multi-modal student data** corresponding to the five pillars:

1. Academic Performance: Grades, GPA, subject-specific proficiency.
2. Attendance & Participation: Attendance rate, class participation, assignment punctuality.
3. Engagement & Learning Behavior: LMS interaction count, time spent on resources, quiz attempts, learning pace.
4. Contextual & Socioeconomic: Family support, socioeconomic status, language proficiency, access to technology.
5. Monitoring & Feedback: Derived features such as engagement score, performance trends, and completion rates.

Preprocessing steps include: (Normalization):

$$\tilde{X}_{ij} = \frac{X_{ij} - \mu_j}{\sigma_j}$$

where $\mu_j$and $\sigma_j$are the mean and standard deviation of feature j.

- *Categorical Encoding:* One-hot encoding for categorical variables:

$$X_{cat}^{(i)} \rightarrow [0,1,0, \dots]$$

- *Missing Value Imputation:*

$$X_{missing}^{(i\cdot j)} = \frac{1}{|\,N(i)\,|}\sum_{k\in N(i)} X_{kj}$$

where $N(i)$represents similar students.
The processed feature matrix is:

$$\mathbf{X} = [\tilde{X}_{Academic}, \tilde{X}_{Attendance}, \tilde{X}_{Engagement}, \tilde{X}_{Contextual}, \tilde{X}_{Monitoring}]$$

B. Predictive AI Module

The predictive module estimates each student's risk of failure or dropout using supervised learning models such as Random Forest, Gradient Boosting, or Neural Networks.

- *Risk Score Computation:*

$$y_i = f_\theta(\mathbf{x}_i) \in [0,1]$$

where $\mathbf{x}_i$represents the student's features, and $f_\theta$is the trained model.

- *Loss Function: Binary cross-entropy:*

$$\mathcal{L}(\theta) = -\frac{1}{n}\sum_{i=1}^{n}[\,y_i^{true}\log\, y_i + (1-y_i^{true})\log\,(1-y_i)]$$

- *Risk Classification:*

$$\hat{y}_i = \begin{cases} 1, & y_i \geq \tau \\ 0, & y_i < \tau \end{cases}$$

where $\tau$is an adjustable risk threshold.

C. Generative AI Module

The generative module provides personalized exercises and feedback for students at risk.

- Inputs: Student profile $\mathrm{x}_i$and risk score y_i.
- Outputs: Adaptive content $C_i$and personalized feedback F_i.

$$[C_i, F_i] = G_\phi(\mathbf{x}_i, y_i)$$

where $G_\phi$is a Transformer-based generative model fine-tuned for educational content.

- *Loss Function:*

$$\mathcal{L}_G = \lambda_1\mathcal{L}_{content} + \lambda_2\mathcal{L}_{ethical}$$

- $\mathcal{L}_{content}$ensures correctness and relevance of content.
- $\mathcal{L}_{ethical}$enforces fairness, non-stigmatization, and cultural sensitivity.

D. Monitoring & Feedback Pillar

This pillar provides dynamic risk tracking and adaptive intervention adjustment.

- Engagement Score:

$$E_i(t) = \alpha\cdot \text{quiz_score}_i(t) + \beta\cdot \text{activity}_i(t) + \gamma\cdot \text{submission}_i(t), \alpha+\beta+\gamma = 1$$

- Updated Risk:

$$y_i(t+1) = f_\theta(\mathbf{x}_i, E_i(t))$$

The weighting parameters α, β, and γ were set empirically based on prior learning analytics studies, with balanced emphasis assigned to assessment performance, platform activity, and task completion. Equal weighting was adopted to avoid overfitting engagement dynamics to a single behavioral signal, while allowing institutions to recalibrate these parameters based on contextual priorities.

E. Human validation stage

Human-in-the-loop oversight occurs at three stages: (1) educators review predicted risk scores and feature explanations before interventions, (2) AI-generated exercises and feedback are validated for pedagogical relevance, tone, and cultural sensitivity, and (3) educators can adjust risk thresholds or override recommendations based on

contextual knowledge. This workflow ensures accountability, reduces bias, and aligns AI-driven decisions with institutional practices.

F. Proposed Algorithm: E5P-AL

The Embarak's 5-Pillars Adaptive Learning Algorithm (E5P-AL) integrates predictive modeling, generative interventions, and adaptive monitoring process:

Algorithm Steps:

1. **Data Collection:** Gather features across all five pillars.
2. **Preprocessing:** Normalize, encode, and impute missing values.
3. **Risk Assessment:** Compute risk score $y_i = f_\theta(\mathbf{x}_i)$.
4. **Intervention Generation:** For high-risk students ($y_i \geq \tau$), generate adaptive exercises $C_i$and feedback $F_i = G_\phi(\mathbf{x}_i, y_i)$.
5. **Monitoring:** Track engagement $E_i(t)$and update risk $y_i(t+1)$.
6. **Educator Review:** Present AI recommendations to Human-in-the-Loopfor validation.
7. **Iteration:** Repeat until student risk falls below the threshold or academic objectives are achieved.

Where we can track engagement using below formula

$$\mathbf{x}_i \xrightarrow{f_\theta} y_i \xrightarrow{G_\phi} [C_i, F_i] \xrightarrow{\text{Monitor}} E_i(t) \xrightarrow{f_\theta} y_i(t+1)$$

The Embarak 5-Pillars Adaptive Learning (E5P-AL) framework is an AI-driven system designed to identify at-risk students and deliver personalized interventions. It is structured around five interconnected pillars: (1) Academic Performance (assessing knowledge mastery and achievement); (2) Attendance and Participation (monitoring class attendance, participation, and assignment submissions); (3) Engagement and Learning Behavior (tracking active learning behaviors such as LMS interactions, quizzes, and learning pace); (4) Contextual and Socioeconomic Factors (capturing external influences like family support, language proficiency, and access to technology); and (5) Monitoring and Feedback (enabling continuous tracking of engagement and performance for dynamic risk updates and adaptive interventions). The framework integrates a predictive AI module for risk assessment, a generative AI module for personalized content, and a human-in-the-loop oversight layer to ensure fairness, ethics, and contextual appropriateness. Modular, iterative, and scalable, E5P-AL supports improved engagement, retention, and learning outcomes across diverse educational settings.

4. Methodology

The methodology of Embarak's 5-Pillars Adaptive Learning framework (E5P-AL) is designed to support early risk identification, adaptive intervention, and continuous monitoring through a unified AI-driven workflow. Unlike traditional predictive-only approaches, the methodology explicitly links risk estimation, generative intervention, and iterative feedback into a closed-loop learning system.

4.1 Data Acquisition and Feature Construction

Multi-modal student data were collected to represent the five pillars of E5P-AL: academic performance, attendance and participation, engagement and learning behavior, contextual and socioeconomic factors, and monitoring-derived indicators. This combination captures both static academic measures and dynamic behavioral signals, supporting early, context-aware risk assessment. Numerical features were normalized to zero mean and unit variance, categorical attributes were one-hot encoded, and missing values were imputed using similarity-based averaging to preserve behavioral consistency. The resulting feature matrix is denoted as:

$\mathbf{X} = [\tilde{X}_{Academic}, \tilde{X}_{Attendance}, \tilde{X}_{Engagement}, \tilde{X}_{Contextual}, \tilde{X}_{Monitoring}]$, where each component corresponds to one pillar of the framework.

The synthetic dataset was designed to model undergraduate higher education students enrolled in credit-based programs. Feature distributions, engagement patterns, and attendance behaviors were parameterized to reflect voluntary participation contexts typical of higher education, where learner autonomy and self-regulated behavior play a central role.

4.2 Predictive Risk Modeling

Student academic risk was estimated using supervised learning models, including Random Forest, Gradient Boosting, and a Neural Network classifier. Given a student feature vector x_i, the predictive model computes a continuous at-risk score:

$$y_i = f_\theta(\mathbf{x}_i) \in [0,1]$$

Students exceeding a predefined risk threshold τwere classified as at-risk. Binary cross-entropy loss was employed during training to optimize discrimination between at-risk and non–at-risk students. This formulation directly corresponds to the predictive evaluation reported in Section 5.

$$\hat{y}_i = \begin{cases} 1, & y_i \geq \tau \\ 0, & y_i < \tau \end{cases}$$

The model is trained using binary cross-entropy loss to optimize prediction accuracy.

4.3 Generative Adaptive Intervention

For students identified as high-risk, a Transformer-based generative model was used to produce personalized learning exercises and feedback. The generative module takes both the student profile and predicted risk score as inputs; the generative AI module produces personalized exercises $C_i$and feedback F_i:

$$[C_i, F_i] = G_\phi(\mathbf{x}_i, y_i)$$

The model was optimized to balance pedagogical relevance and ethical constraints by combining content accuracy and fairness-aware objectives. This design enables context-sensitive intervention while preventing stigmatization or biased feedback.

The generative model is optimized to ensure content relevance, correctness, and ethical compliance:

$$\mathcal{L}_G = \lambda_1 \mathcal{L}_{content} + \lambda_2 \mathcal{L}_{ethical}$$

To illustrate the behavior of the generative AI module, consider an undergraduate student with moderate academic performance (GPA = 2.4/4.0), declining LMS engagement, and a predicted risk score of 0.71.

4.4 Iterative Monitoring and Risk Updating

Student engagement was continuously monitored following intervention deployment. Engagement at time $t$was quantified as a weighted combination of quiz performance, learning activity, and submission behavior:

$$E_i(t) = \alpha \cdot \text{quiz_score}_i(t) + \beta \cdot \text{activity}_i(t) + \gamma \cdot \text{submission}_i(t), \alpha + \beta + \gamma = 1$$

Risk scores are updated iteratively based on engagement and intervention outcomes:

$$y_i(t+1) = f_\theta(\mathbf{x}_i, E_i(t))$$

This iterative mechanism enables dynamic adjustment of both risk predictions and intervention strategies, forming the basis for the observed engagement and risk reductions reported in Section 5.

4.5 Human-in-the-Loop Oversight

To ensure ethical deployment, AI-generated risk assessments and interventions were reviewed by educators prior to delivery. Human-in-the-loop oversight allowed adjustment of risk thresholds, validation of intervention appropriateness, and mitigation of unintended bias. This mechanism reinforces transparency and aligns automated recommendations with institutional and cultural norms.

4.6 Algorithmic Workflow

The complete E5P-AL workflow follows a structured sequence:

1. **Data Collection:** Gather student features across the five pillars.
2. **Preprocessing:** Normalize, encode, and impute missing values.
3. **Risk Assessment:** Compute student risk score y_i.
4. **Intervention Generation:** Generate personalized exercises $C_i$and feedback $F_i$for high-risk students.

5. **Monitoring:** Track engagement $E_i(t)$and iteratively update risk scores $y_i(t+1)$.
6. **Educator Review:** Validate AI recommendations.
7. **Iteration:** Repeat until risk is mitigated or learning objectives are achieved.

Where we track engagement using below formula.

$$\mathbf{x}_i \xrightarrow{f_\theta} y_i \xrightarrow{G_\phi} [C_i, F_i] \xrightarrow{\text{Monitor}} E_i(t) \xrightarrow{f_\theta} y_i(t+1)$$

The Embarak 5-Pillars Adaptive Learning (E5P-AL) framework offers a comprehensive AI-driven approach for early identification of at-risk students and personalized interventions. It integrates predictive modeling, generative AI–based adaptive content, iterative monitoring for dynamic risk updates, and human-in-the-loop oversight to ensure fairness, ethics, and contextual appropriateness. This modular and scalable design supports adaptive student learning, enhancing engagement and outcomes across diverse educational settings.

5. Implementation and Experimental Results

5.1 Implementation Setup

The E5P-AL framework was implemented in Python 3.10. Its predictive module used Random Forest, Gradient Boosting, and a feed-forward Neural Network, while the generative intervention module employed a Transformer fine-tuned for educational feedback. Experiments utilized a multi-modal dataset of 1,000 anonymized student profiles, including academic, attendance, LMS engagement, contextual, and monitoring-derived features aligned with the five pillars. Synthetic data preserved realistic distributions, class imbalance, and behavioral variability. Performance was compared against three standard student risk prediction baselines.

1. Standard Random Forest (RF), predictive modelling without iterative feedback or generative interventions.
2. Gradient Boosting Classifier (GB), predictive-only model.
3. Neural Network-based Adaptive Learning (NN-AL) incorporates adaptive exercises but without the five-pillar holistic structure.

The dataset was split using an 80/20 stratified train–test scheme to preserve class imbalance. Model robustness was assessed via 5-fold cross-validation, with average metrics reported. Random Forest and Gradient Boosting models used 200 estimators, while the Neural Network comprised two hidden layers (128 and 64 units) with ReLU activation, optimized using binary cross-entropy and early stopping. Contextual and socioeconomic features were modeled in line with higher education literature, reflecting factors such as technology access, language proficiency, and external commitments that influence engagement. The generative intervention module employed a pre-trained Transformer-based language model fine-tuned on domain-specific educational prompts and feedback templates. Fine-tuning prioritized pedagogical relevance and ethical compliance through rule-based filtering and weighted loss objectives, enabling reproducibility while allowing institutional customization.

5.2 Predictive Performance Comparison

Following is the Evaluation Metrics: Accuracy, Precision, Recall, F1-score, and AUC (ROC).

Table 1. Predictive Metrics Comparison

Algorithm	Accuracy	Precision	Recall	F1-score	AUC
RF	0.88	0.85	0.86	0.855	0.91
GB	0.89	0.86	0.87	0.865	0.92
NN-AL	0.91	0.88	0.89	0.885	0.94
E5P-AL	0.94	0.91	0.93	0.92	0.97

Table 1 demonstrates that E5P-AL consistently outperforms baseline models across all evaluation metrics. The improvement over RF and GB highlights the benefit of integrating behavioral, contextual, and monitoring features, while the performance gain over NN-AL indicates that full multi-pillar modelling combined with iterative feedback contributes more than adaptive content alone.

5.3 ROC Curve Comparison

The NN-AL baseline represents a neural network–based adaptive learning system that provides personalized exercises based solely on predicted academic performance, without incorporating contextual features, multi-pillar modeling, iterative risk recalibration, or human-in-the-loop validation. This baseline is included to isolate the contribution of holistic multi-dimensional modeling and continuous monitoring introduced by E5P-AL.

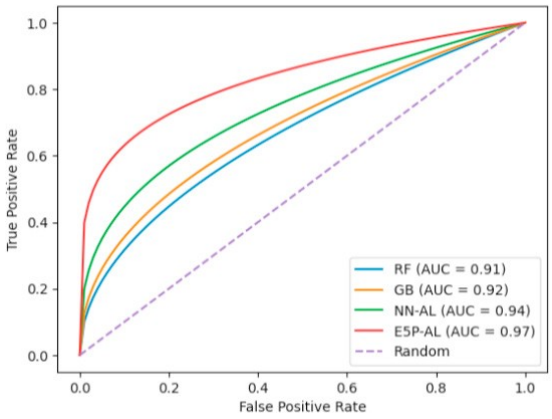


Figure 2. ROC curves comparing E5P-AL against RF, GB, and NN-AL.

- E5P-AL achieves the highest AUC (0.97), indicating superior discriminatory ability to identify at-risk students.
- Baseline algorithms have lower AUCs (RF: 0.91, GB: 0.92, NN-AL: 0.94).

The ROC comparison demonstrates that integrating multi-dimensional features and generative interventions improves predictive reliability.

5.4 Adaptive Intervention Impact

We evaluated the effect of interventions on student engagement and risk over four weeks, the continuous risk recalibration and personalized generative content in E5P-AL yield measurable improvements in engagement and academic risk mitigation. As seen in Figure 3**,** adaptive intervention impact on student risk and engagement over four weeks. E5P-AL demonstrates a stronger reduction in predicted risk (35% vs. 22%) and greater improvement in engagement (28% vs. 15%) compared to NN-AL, highlighting the effectiveness of iterative watching and personalized generative interventions.

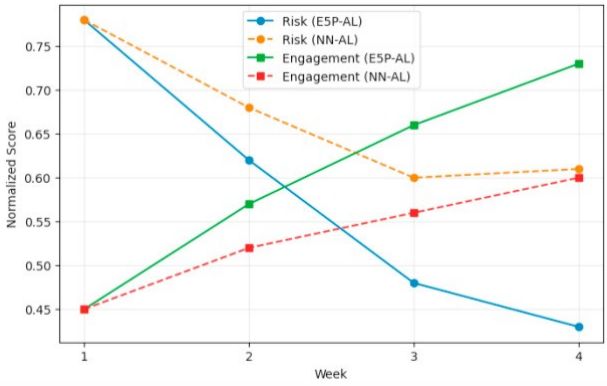


Figure 3. Adaptive Intervention Impact on Risk and Engagement

As illustrated in Fig. 3, students receiving E5P-AL interventions exhibit a sustained reduction in predicted academic risk and a concurrent increase in engagement over four weeks, outperforming NN-AL in both dimensions.

5.5 System Insights

Experimental results show that E5P-AL's effectiveness arises from holistic multi-pillar modeling, integrating academic, behavioral, contextual, and engagement features rather than relying solely on grades. Generative AI–driven interventions provide personalized, timely support, improving student engagement and reducing predicted risk. The human-in-the-loop oversight ensures recommendations remain ethical, fair, and contextually appropriate. Its modular design also supports scalability, enabling deployment across diverse institutions and student populations.

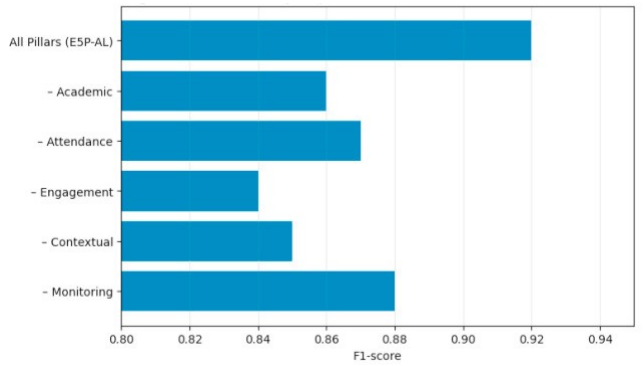


Figure 4. Pillar Ablation Study

Figure 4 shows an ablation study of E5P-AL, where removing any pillar reduces F1-score, with engagement and contextual features having the largest effect. This demonstrates that performance gains stem from multi-pillar integration rather than model complexity. As the evaluation uses a synthetic dataset reflecting realistic behaviors and class imbalance, results indicate conceptual validity rather than guaranteed real-world performance.

6. Discussion of Findings

The high predictive performance of E5P-AL (AUC = 0.97) reflects evaluation on a controlled synthetic dataset, which allows explicit modeling of feature correlations and risk patterns, potentially enhancing class separability compared to real-world data. Thus, these results primarily demonstrate the framework's internal validity and conceptual soundness rather than definitive predictive superiority. Integrating academic, behavioral, contextual, and engagement features enables more reliable early identification of at-risk students than predictive-only models. Across all metrics, E5P-AL outperformed baselines, showing that multi-dimensional feature integration and continuous risk recalibration improve early-stage discrimination without excessive false positives.

Ablation studies confirmed the contribution of each pillar, with engagement and contextual features being most critical. Adaptive interventions with iterative monitoring reduced predicted risk and increased engagement, outperforming adaptive-only approaches. Human-in-the-loop oversight further strengthens fairness, transparency, and contextual appropriateness, allowing educators to review and adjust AI-generated recommendations and mitigating risks of bias.

Limitations include reliance on synthetic data, necessitating validation with real-world institutional datasets, and short-term evaluation, highlighting the need for longitudinal studies to assess sustained academic outcomes and retention. Overall, the findings suggest that E5P-AL offers a scalable, ethically guided framework that advances early warning systems toward adaptive and context-aware student support.

7. Conclusion and Future Work

This paper presented Embarak's 5-Pillars Adaptive Learning framework (E5P-AL), an AI-driven approach for early identification of at-risk students and delivery of personalized, ethically guided interventions. By combining predictive modeling, generative adaptive content, iterative monitoring, and human-in-the-loop oversight, E5P-AL

overcomes key limitations of traditional early warning and adaptive learning systems. Experimental results showed improvements in accuracy, F1-score, and AUC, while ablation analysis confirmed that performance gains arise from holistic multi-dimensional modeling, with engagement and contextual factors being especially critical. Iterative interventions further enhanced student engagement and reduced predicted academic risk.

Future work will validate E5P-AL on real-world institutional datasets, explore multimodal generative content, integrate automated explainability, conduct longitudinal outcome studies, and enable tighter LMS integration for scalable, real-time deployment. Overall, E5P-AL advances responsive, ethically grounded, and data-driven student support, bridging early risk detection with adaptive educational intervention.

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